

Sean Luka Girgis

Senior Observability / APM Engineer

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Professional Summary

Senior Observability, APM, and Data Engineering professional with 20+ years of enterprise IT experience across monitoring platforms, performance engineering, telemetry analytics, capacity planning, and automation. Strong fit for enterprise observability roles requiring Dynatrace/AppMon, synthetic monitoring, transaction tracing, dashboards, alerting, Python scripting, Linux/Unix, documentation, performance analysis, and stakeholder reporting. Experienced turning large-scale infrastructure and application telemetry into actionable KPIs, forecasts, runbooks, and executive-ready insights in regulated enterprise environments.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas telemetry pipelines ingesting P95 performance and capacity data from 6,000+ infrastructure endpoints, replacing manual reporting with repeatable, validated workflows.
- Managed and integrated enterprise monitoring feeds from BMC TrueSight/TSCO, CA Wily/APM, and AppDynamics into unified reporting datasets for capacity planning, performance analysis, and executive risk visibility.
- Developed dashboards and operational reports using Streamlit, matplotlib, seaborn, and plotly to surface infrastructure KPIs, trends, bottlenecks, and capacity risks for senior stakeholders.
- Applied performance and capacity analysis across CPU, memory, utilization, and infrastructure telemetry to identify underutilized assets, forecast demand, and support cost-aware optimization.
- Designed data quality, reconciliation, and validation checks for telemetry pipelines to improve trust in monitoring reports, forecasts, and operational decision-making.
- Created ML forecasting models with Prophet and scikit-learn to predict capacity bottlenecks months ahead and support proactive remediation planning.
- Supported operational excellence through troubleshooting, root-cause analysis, documentation, stakeholder communication, and repeatable runbook-style practices.
- Worked in a regulated banking environment with audit, access, compliance, and data-governance expectations for infrastructure and application reporting.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting production observability across web and mobile platforms.
- Implemented End-User Monitoring and end-to-end transaction tracing to improve visibility into user experience, response times, and application performance behavior.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and deliver actionable recommendations to engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior with APM telemetry and support performance-driven release decisions.

- Upgraded Dynatrace AppMon from 6.5 to 7.0 and enforced TLS 1.2 security without regression in production monitoring coverage.
- Produced performance reports, dashboards, and technical findings for engineering and business stakeholders.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large enterprise environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed and maintained dashboards, alerts, SLA thresholds, management modules, and performance reporting standards used across client monitoring environments.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and reduce manual weekly export work.
- Provided sizing recommendations, agent onboarding guidance, and Golden Image deployment patterns to improve monitoring consistency and reduce setup time.
- Collaborated with J2EE/WebLogic performance teams to troubleshoot bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure signals.
- Delivered client training, technical documentation, and operational guidance for monitoring standards, agent deployments, and production troubleshooting.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits and bottlenecks across JDBC connections, threads, heap memory, CPU, and garbage collection behavior.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve visibility during performance tests.
- Documented performance baselines and regression thresholds used by engineering teams for release gate decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving strict transaction performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while maintaining sub-second query latency.
- Built validation tooling to verify data correctness and performance behavior across large transaction datasets during phased migration.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

Enterprise Capacity & Telemetry Analytics Platform

Built telemetry pipelines and dashboards that consolidated infrastructure monitoring data from thousands of endpoints into KPIs, forecasts, and executive-ready capacity insights.

Technologies: Python, Pandas, BMC TrueSight/TSCO, CA APM, AppDynamics, Oracle, SQL, Streamlit

HorizonScale — AI Capacity Forecasting Engine

Developed a capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to predict infrastructure bottlenecks and support proactive planning.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Data, Forecasting

FAST Project

Mined real-user and synthetic performance metrics from Dynatrace and Gomez Synthetic Monitoring to identify performance improvements for critical production systems.

Technologies: Dynatrace, Gomez Synthetic Monitoring, EUM, Transaction Tracing, Performance Engineering

AI-Powered Job Search Pipeline

Built a Python-based agentic workflow with semantic duplicate detection, LLM scoring, JSON artifact generation, quality gates, and tracking, demonstrating practical automation and AI-assisted development patterns.

Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG, JSON Contracts, Automation